Alfido Tech Cybersecurity Internship

Submitted By

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Organization: Alfido Tech
Task: Task 1 – Reconnaissance & Vulnerability Scanning
Tools Used: RustScan, Nmap, Ubuntu WSL
Date: May 2026

1. Objective

The objective of this task was to perform reconnaissance and vulnerability scanning on a target system using cybersecurity scanning tools. The task focused on identifying:

- Open ports
- Running services
- Service versions
- Potential vulnerabilities
- Security misconfigurations

This task helps in understanding attack surfaces and the importance of secure service configurations in a network environment.

2. Environment Setup

Operating Environment

- Ubuntu 24.04 on Windows Subsystem for Linux (WSL)

Tools Used

- RustScan 2.4.1
- Nmap 7.94SVN

3. Tool Installation

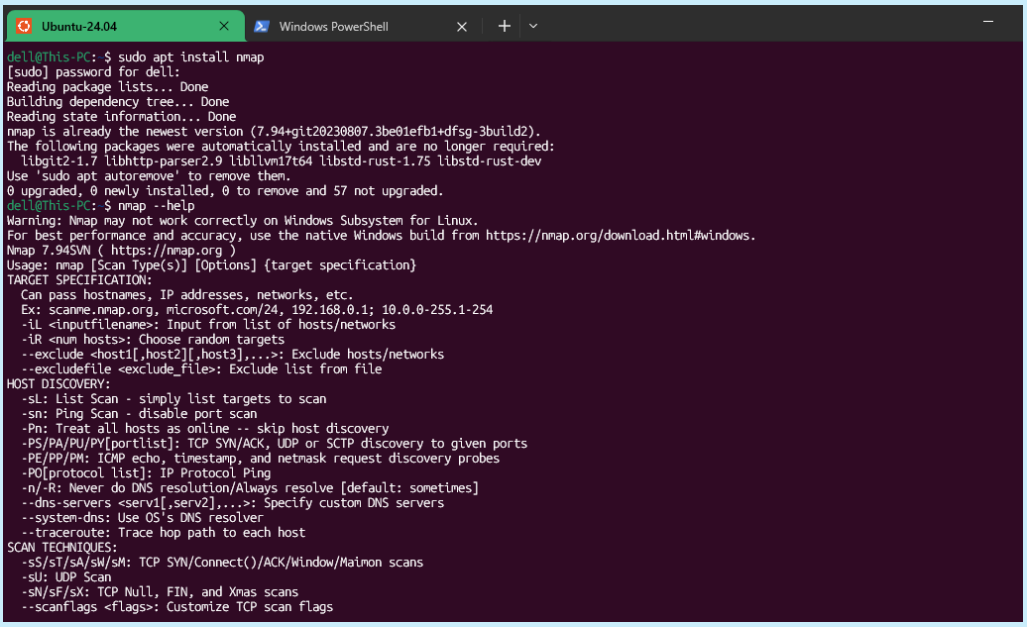
The following commands were used to install and verify the required tools.

Nmap Installation

sudo apt install nmap

Nmap Help Verification

nmap --help

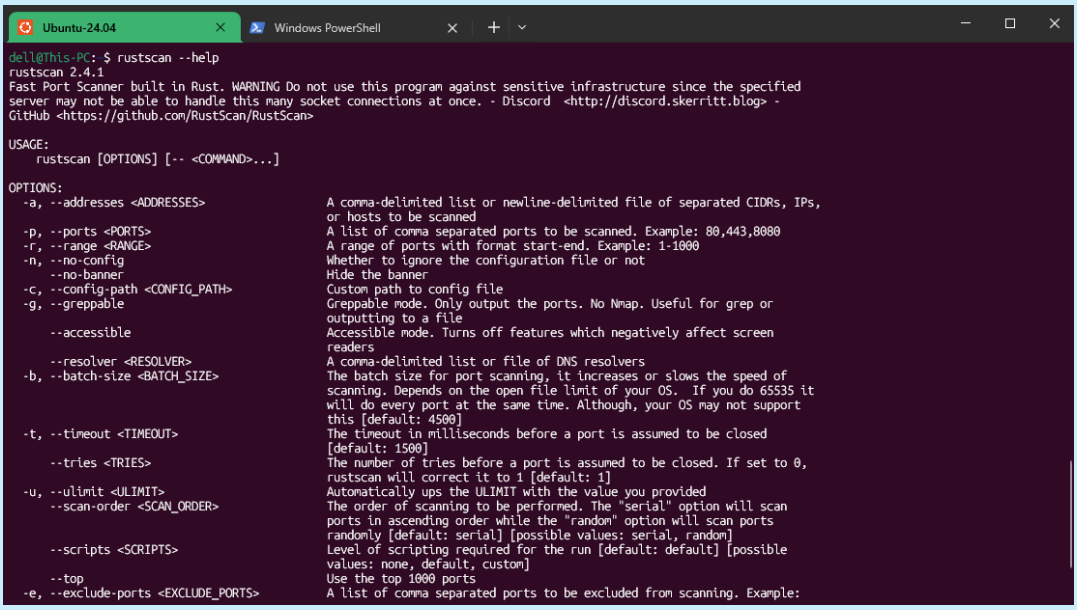


```
Ubuntu-24.04    Windows PowerShell

dell@This-PC: $ sudo apt install nmap
[sudo] password for dell:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
nmap is already the newest version (7.94+git20230807.3be01efb1+dfsg-3build2).
The following packages were automatically installed and are no longer required:
  libgit2-1.7 libhttp-parser2.9 libllvm17t64 libstd-rust-1.75 libstd-rust-dev
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 57 not upgraded.
dell@This-PC: $ nmap --help
Warning: Nmap may not work correctly on Windows Subsystem for Linux.
For best performance and accuracy, use the native Windows build from https://nmap.org/download.html#windows.
Nmap 7.94SVN ( https://nmap.org )
Usage: nmap [Scan Type(s)] [Options] {target specification}
TARGET SPECIFICATION:
  Can pass hostnames, IP addresses, networks, etc.
  Ex: scanme.nmap.org, microsoft.com/24, 192.168.0.1; 10.0.0-255.1-254
  -iL <inputfilename>: Input from list of hosts/networks
  -iR <num hosts>: Choose random targets
  --exclude <host1[,host2][,host3],...>: Exclude hosts/networks
  --excludefile <exclude_file>: Exclude list from file
HOST DISCOVERY:
  -sL: List Scan - simply list targets to scan
  -sn: Ping Scan - disable port scan
  -Pn: Treat all hosts as online -- skip host discovery
  -PS/PA/PU/PY[portlist]: TCP SYN/ACK, UDP or SCTP discovery to given ports
  -PE/PP/PM: ICMP echo, timestamp, and netmask request discovery probes
  -PO[protocol list]: IP Protocol Ping
  -n/-R: Never do DNS resolution/Always resolve [default: sometimes]
  --dns-servers <serv1[,serv2],...>: Specify custom DNS servers
  --system-dns: Use OS's DNS resolver
  --traceroute: Trace hop path to each host
SCAN TECHNIQUES:
  -sS/sT/sA/sW/sM: TCP SYN/Connect()/ACK/Window/Maimon scans
  -sU: UDP Scan
  -sN/sF/sX: TCP Null, FIN, and Xmas scans
  --scanflags <flags>: Customize TCP scan flags
```

RustScan Help Verification

rustscan --help



```
Ubuntu-24.04    Windows PowerShell

dell@This-PC: $ rustscan --help
rustscan 2.4.1
Fast Port Scanner built in Rust. WARNING Do not use this program against sensitive infrastructure since the specified
server may not be able to handle this many socket connections at once. - Discord  <http://discord.skerritt.blog> -
GitHub <https://github.com/RustScan/RustScan>

USAGE:
    rustscan [OPTIONS] [-- <COMMAND>...]

OPTIONS:
  -a, --addresses <ADDRESSES>                A comma-delimited list or newline-delimited file of separated CIDRs, IPs,
                                             or hosts to be scanned
  -p, --ports <PORTS>                        A list of comma separated ports to be scanned. Example: 80,443,8080
  -r, --range <RANGE>                        A range of ports with format start-end. Example: 1-1000
  -n, --no-config                            Whether to ignore the configuration file or not
      --no-banner                            Hide the banner
  -c, --config-path <CONFIG_PATH>            Custom path to config file
  -g, --greppable                            Greppable mode. Only output the ports. No Nmap. Useful for grep or
                                             outputting to a file
      --accessible                           Accessible mode. Turns off features which negatively affect screen
                                             readers
      --resolver <RESOLVER>                  A comma-delimited list or file of DNS resolvers
  -b, --batch-size <BATCH_SIZE>              The batch size for port scanning, it increases or slows the speed of
                                             scanning. Depends on the open file limit of your OS.  If you do 65535 it
                                             will do every port at the same time. Although, your OS may not support
                                             this [default: 4500]
  -t, --timeout <TIMEOUT>                    The timeout in milliseconds before a port is assumed to be closed
                                             [default: 1500]
      --tries <TRIES>                        The number of tries before a port is assumed to be closed. If set to 0,
                                             rustscan will correct it to 1 [default: 1]
  -u, --ulimit <ULIMIT>                      Automatically ups the ULIMIT with the value you provided
      --scan-order <SCAN_ORDER>              The order of scanning to be performed. The "serial" option will scan
                                             ports in ascending order while the "random" option will scan ports
                                             randomly [default: serial] [possible values: serial, random]
      --scripts <SCRIPTS>                    Level of scripting required for the run [default: default] [possible
                                             values: none, default, custom]
      --top                                  Use the top 1000 ports
  -e, --exclude-ports <EXCLUDE_PORTS>        A list of comma separated ports to be excluded from scanning. Example:
```

4. Reconnaissance Process

The target system IP address scanned during the assessment: **10.216.3.29**

Port Scanning Using RustScan

Basic Service Version Detection

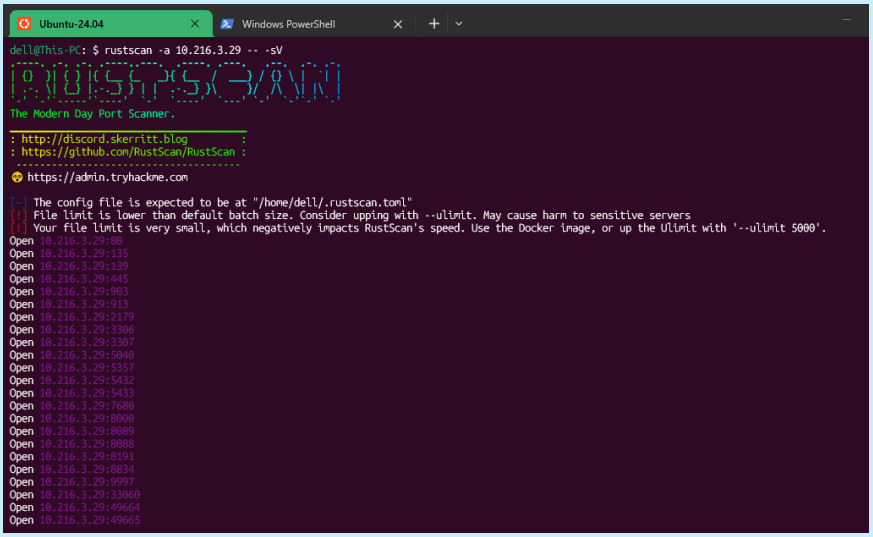
The following command was executed:

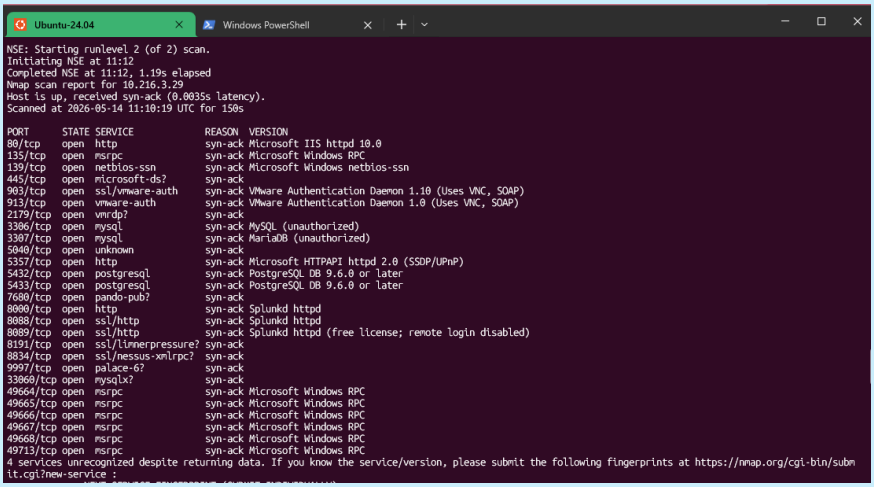
rustscan -a 10.216.3.29 -- -sV

Purpose

This command performs:

- Fast port scanning using RustScan
- Service version detection using Nmap





Findings

Several open ports and services were identified including:

- Microsoft IIS Web Server
- MySQL Database
- MariaDB
- PostgreSQL
- Splunk Services
- VMware Authentication Services
- Windows RPC Services

Open Ports Identified

Port	Service
80	HTTP (Microsoft IIS 10.0)
135	Microsoft RPC
139	NetBIOS
445	SMB
3306	MySQL
3307	MariaDB
5432	PostgreSQL
8000	Splunk HTTP
8088	Splunk SSL HTTP
8089	Splunk Management
8834	Nessus Service

Security Observation

The target system exposed multiple network services publicly, increasing the potential attack surface.

Aggressive Enumeration Scan

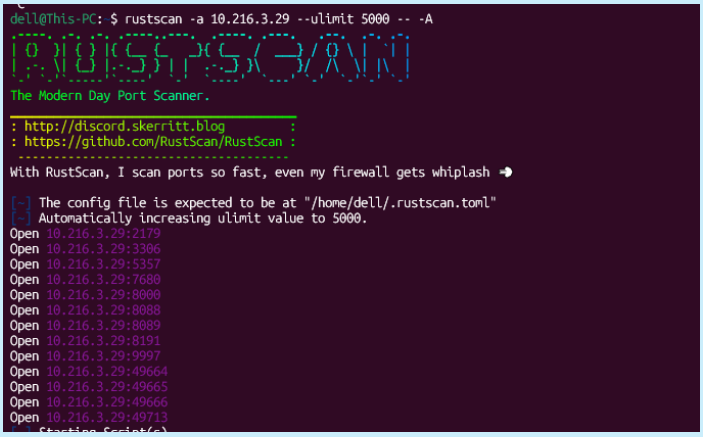
The following command was executed:

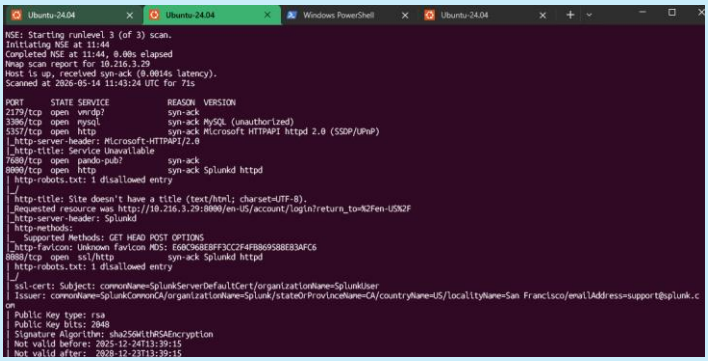
rustscan -a 10.216.3.29 --ulimit 5000 -- -A

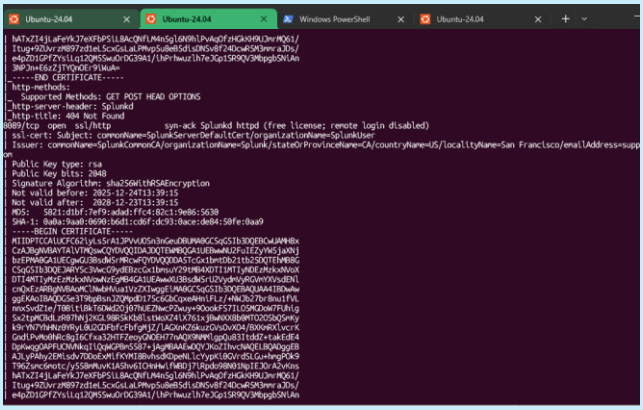
Purpose

This command performs:

- Aggressive service enumeration
- OS detection
- Script scanning
- Traceroute
- Advanced fingerprinting







Findings

The scan revealed:

- Windows-based operating system
- IIS web server running on port 80
- Splunk management interfaces
- Nessus vulnerability management service
- SSL certificates and service metadata

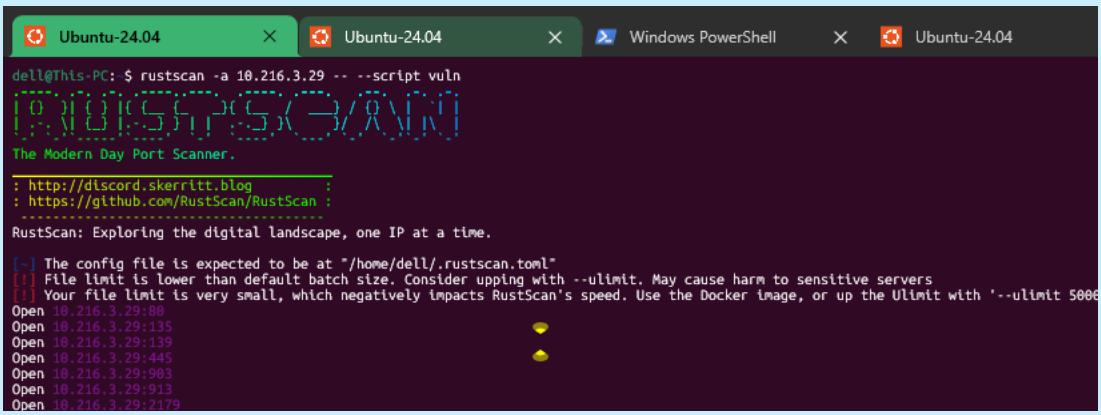
Vulnerability Scanning

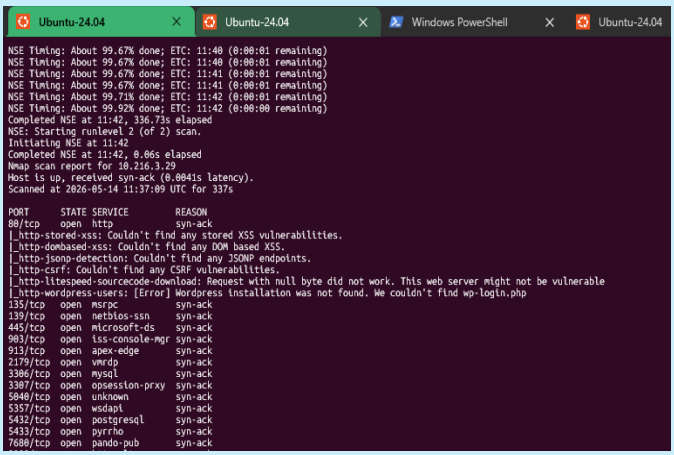
The following command was executed:

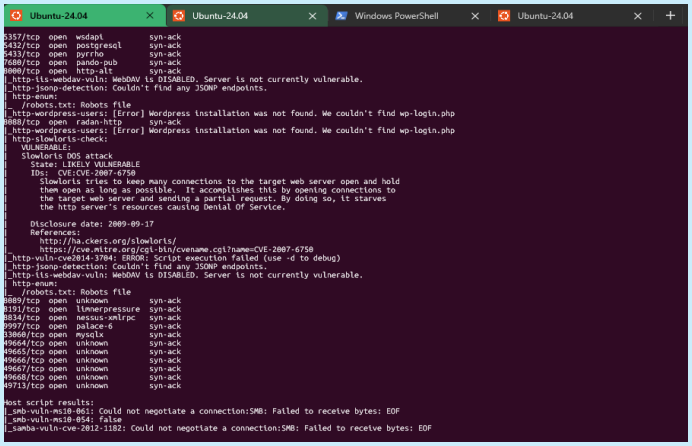
rustscan -a 10.216.3.29 -- --script vuln

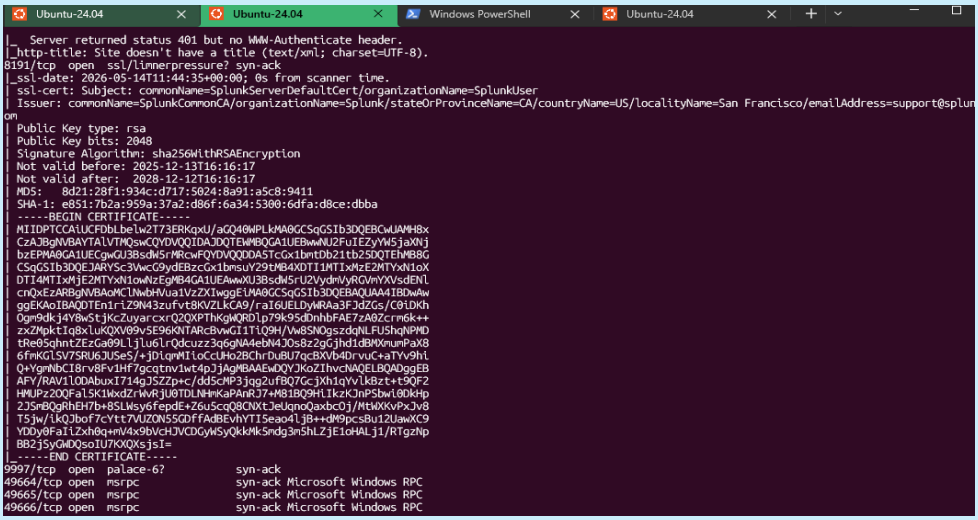
Purpose

To identify known vulnerabilities and weak configurations using Nmap NSE vulnerability scripts.









Vulnerability Findings

Slowloris Denial of Service Vulnerability

The scan identified a potential vulnerability:

Vulnerability	CVE
Slowloris DOS Attack	CVE-2007-6750

Description

Slowloris is a denial-of-service attack that keeps many HTTP connections open simultaneously, exhausting server resources and potentially making the server unavailable to legitimate users.

Risk Level

Medium

5. SMB Enumeration Results

The SMB-related services running on ports 139 and 445 were identified during the scan.

Observation

Some SMB vulnerability checks failed due to negotiation limitations; however, SMB exposure itself presents a security concern because SMB services are common targets for:

- Unauthorized file access
- Lateral movement
- SMB exploitation attacks

6. HTTP & SSL Enumeration

The scan identified:

- HTTP services
- SSL-enabled services
- Splunk SSL certificates

Additional Findings

- Supported HTTP methods were detected
- Robots.txt files were identified
- SSL certificate metadata was visible

7. Key Security Risks Identified

Risk	Description
Open SMB Services	May allow unauthorized access or exploitation
Database Exposure	MySQL and PostgreSQL services exposed
Splunk Exposure	Monitoring dashboards accessible over network
HTTP Services	Potential web attack surface
Slowloris Vulnerability	Possible denial-of-service risk

8. Recommendations

The following recommendations are suggested to improve security posture:

1. Disable unnecessary services and ports.
2. Restrict SMB access using firewall rules.
3. Limit database exposure to internal networks only.
4. Apply latest security patches and updates.
5. Enable intrusion detection and monitoring.
6. Configure rate limiting to reduce DOS attack risks.
7. Restrict administrative services from public access.

9 Conclusion

This task successfully demonstrated the use of RustScan and Nmap for reconnaissance and vulnerability assessment. Multiple open ports, services, and potential vulnerabilities were identified on the target system.

The activity provided practical exposure to:

- Port scanning
- Service enumeration
- Vulnerability identification
- Security analysis
- Attack surface assessment

This exercise improved understanding of real-world reconnaissance techniques and the importance of proactive security hardening.

10. Screenshots Included

The following evidence screenshots were captured during the assessment:

- RustScan help and verification
- Nmap verification
- Service version detection scan
- Vulnerability scan results
- SSL certificate enumeration
- Aggressive scan results
- Open ports and services output